

Achieving Better Category Separability for Hyperspectral Image Classification: A Spatial–Spectral Approach

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Abstract—The task of hyperspectral image (HSI) classification has attracted extensive attention. The rich spectral information in HSIs not only provides more detailed information but also brings a lot of redundant information. **Redundant information makes spectral curves of different categories have similar trends, which leads to poor category separability.** In this article, we achieve better category separability from the perspective of **increasing the difference between categories** and **reducing the variation within category**, thus improving the classification accuracy. Specifically, **we propose the template spectrum-based processing module from spectral perspective**, which can effectively expose the unique characteristics of different categories and reduce the difficulty of model mining key features. Second, **we design an adaptive dual attention network from spatial perspective**, where the target pixel can adaptively aggregate high-level features by evaluating the confidence of effective information in different receptive fields. Compared with the single adjacency scheme, the adaptive dual attention mechanism makes the ability of target pixel to combine spatial information to reduce variation more stable. Finally, **we designed a dispersion loss from the classifier's perspective.**

By supervising the learnable parameters of the final classification layer, the loss makes the category standard eigenvectors learned by the model more dispersed, which improves the category separability and reduces the rate of misclassification. Experiments on three common datasets show that our proposed method is superior to the comparison method.

Index Terms—Category separability, dispersion loss, dual attentive mechanism, hyperspectral image classification (HSIC), template spectrum.

I. INTRODUCTION

HYPERSPECTRAL images (HSIs) contain rich spectral information due to hundreds of continuous and narrow spectral bands [1], [2]. The pixel-level HSI classification (HSIC) with promised accuracy facilitates emerging applications in various fields, such as resource exploration, disaster prevention and rescue, and urban planning [3], [4], [5], [6], [7]. The performance of HSIC relies on the category separability, which refers to the degree of differentiation between categories. The better category separability helps to reduce the difficulty of the model to mine the key features that distinguish different categories, thereby leading to better classification results.

Achieving better category separability for HSI needs to properly tackle two issues.

- 1) The common trend of spectra in different categories. HSIs contain lots of redundant information in the spectral dimension. Such redundant information makes the overall bias information existing in the HSIs, which is manifested by the similar trend in the spectral curves of different categories.
- 2) The high variability of samples within the same category. The spectrum of the samples contains not only the universal characteristics of the corresponding categories but also their own unique characteristics.

The large fluctuations between the spectral curves of the pixels belonging to the same category serve as an indication. This results in overlapping areas between the distribution ranges of multiple categories, making it more challenging to distinguish samples from different categories. Fig. 1 presents the spectral curves of all samples in three categories on the Indian Pines (IP) dataset. It shows that the curves in different categories exhibit a similar trend, while curves within

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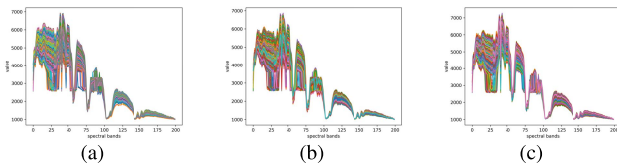


Fig. 1. Spectral curves of three categories in the IP dataset. (a)–(c) Spectral curves of the Corn-Notill category, Corn-mintill category, and Corn category, respectively.

each category exhibit significant fluctuations. The HSIs have poor category separability due to the common trend and high variability. These two factors make the HSI as a challenging task to distinguish different categories, thereby hindering the improvement of classification accuracy.

Existing works focus on enhancing the category separability of HSIs from both spectral and spatial perspectives. From the spectral perspective, it can effectively increase the difference between different categories and improve the category separability by removing the redundant information in the spectra. Both feature extraction (FE) and band selection (BS) are widely used techniques to remove redundant information from the spectra. The principal component analysis (PCA) [8] is a commonly used FE method, which reduces the number of dimensions by mapping the spectra. PCA produces good results, but the physical significance of spectral bands is lost in the process. BS methods compose new spectra by selecting subsets from the original spectral bands, which have better interpretability. However, the FE and BS methods run the risk of losing information that is required for the classification task because they both rely on expert experience to determine the number of retained spectral bands. It is important to emphasize that the existing methods all aim to remove redundant information by reducing the number of spectral bands and do not consider the common trend caused by the overall bias.

From the spatial perspective, the target pixel (pixel to be classified) can effectively reduce its own variability by combining spatial neighborhood information, which can lessen the overlap of various category distributions and thus enhance the category separability. In the spatial neighborhood, pixels in the same category as the target pixel provide effective information that helps the target pixel reduce its own variance, while those in different categories from the target pixel provide interference information. The spectrum combining the effective information fits more closely to the standard category spectrum than the original one. Therefore, the major strategy for reducing the variability is to provide the target pixel more effective information. Several methods make the spatial neighborhood contain more effective information by designing the measurement of adjacent pixels. For example, the spatial-spectral Euclidean distance measurement [9] and the kernel spectral angle mapper (KSAM)-spectral information divergence (SID) measurement [10] are designed to choose adjacent pixels that are more likely to belong to the same category as the target pixel. These methods require additional computation and expert experience to balance the weights among the submetrics. In addition, several works [11], [12], [13], [14], [15] provide more effective information for target

pixel by using superpixels to expand the receptive field. However, these methods rely too much on the segmentation results of the superpixels, and the superpixels lose some detailed feature information compared with the original pixels.

In summary, despite the existing efforts, achieving better category separability for HSI is facing the following challenging problems.

- 1) It is challenging to effectively remove the spectral redundant information from HSIs with less manual experience. Both FE and BS methods aim at removing redundant information by shortening the spectral dimension. For different datasets, the number of retained spectral bands usually needs to be set individually by expert experience according to the characteristics of the dataset. Too much retention of spectral bands will weaken the effect of redundancy removal, and too little retention will increase the risk of losing useful information.
- 2) It is challenging to integrate the target pixel (pixel to be classified) with spatial effective information more efficiently to decrease its own variability. The spatial neighborhood range of the target pixel is restricted due to the computational cost constraint and the difference in the location of the target pixel, which makes the spatial information available to the target pixel limited. At the same time, spatial information inevitably carries interference information, which introduces features of other categories for the target pixel and thus impairs the correct classification of the target pixel.

In this article, we strive to achieve better category separability for HSI via addressing the abovementioned challenges. To that end, we propose a spatial-spectral approach, that is, a template spectrum-based approach from the spectral perspective and the adaptive dual attention network (ADAN) with dispersion loss from the spatial perspective. Specifically, first, we extract the template spectra representing the overall bias information from the HSIs, which can reflect the common trend of the spectral curves of various categories. By removing the template spectrum from the original spectra, the key features of various categories are exposed, which effectively increases the differences between categories. Second, the ADAN allows the target pixel to filter effective information from original pixels and superpixels for aggregation and adaptively fuse high-level features by evaluating the confidence of effective information in the spatial neighborhood. This effectively reduces the variability of the target pixel. Finally, the dispersion loss disperses the standard eigenvectors of different categories as much as possible by supervising the learnable parameters of the fully connected classification layer. This further reduces the overlap of the distributions of the different categories and thus reduces the possibility of misclassification.

The main contributions of this article are summarized as follows.

- 1) We propose the template spectrum-based processing module from a spectral perspective to expose the differences between various categories. The processing can easily and effectively extract the template spectrum containing overall bias information from HSIs. When

the template spectrum is removed from the original spectra, the typical and unique characteristics of the spectrum belonging to different categories are exposed, which reduces the difficulty of model learning to distinguish key features of various categories. The processing removes the redundant information between the spectra from the perspective of overall bias while preserving the original spectral dimension and reducing the dependence on manual.

- 2) We design an ADAN from a spatial perspective to make the target pixel acquire effective information more efficiently and reduce its own variability. In this network, the target pixel obtains the effective information from the adjacent original pixels and the adjacent superpixels. Also, the target pixel can adaptively fuse high-level features by evaluating the confidence of effective information in the neighborhood. The two different receptive fields provide more effective information for target pixel, and the adaptive filtering mechanism reduces the influence of interference information.
- 3) We design the dispersion loss from the perspective of classifier to make the standard eigenvectors of various categories disperse as much as possible by supervising the learnable parameters of the fully connected classification layer. This loss reduces the overlap of sample distributions of different categories by reducing the similarity of standard eigenvectors of different categories, which improves the category separability and reduces the possibility of misclassification.

The remaining sections of this article are structured as follows. Section II examines relevant works. The specifics of the proposed approach are then covered in Section III. The experimental setup and results analysis are introduced in Section IV. This article is concluded in Section V.

II. RELATED WORKS

A. Dimensionality Reduction Methods

There are two main methods to reduce spectral dimension, FE, and BS. FE methods convert original hyperspectral data to low-dimensional data by linear or nonlinear mapping. For example, Rodarmel and Shan [8] used the PCA to extract features from HSIs. In [16], the independent component analysis (ICA) is used to extract features in an unsupervised way. Bandos et al. [17] designed a supervised FE method based on linear discriminant analysis (LDA). In addition, some works are devoted to using manifold learning algorithms to learn low-dimensional features using the underlying geometry of hyperspectral data, such as Laplacian eigenmaps (LEs) [18], local linear embedding (LLE) [19], and Isomap [20].

BS methods are a research hotspot in recent years. Some ranking-based works [21], [22], [23] are aimed at finding an effective criterion to measure the significance of each spectral band and then selecting the top bands to form a new spectrum. The searching-based approach [24], [25] seeks for the smallest subset of feasible bands by adding or deleting spectral bands from the candidate set. Clustering-based methods [26], [27], [28], [29] divide spectral bands into several clusters through

clustering algorithm and then select the most representative band from each cluster.

B. Hyperspectral Image Classification Methods

Over the past decade, many approaches based on machine learning have been proposed, for example, random forest, support vector machine (SVM) [30], and K-nearest neighbor (KNN) [31]. With the development of neural network research, many methods based on neural networks have been proposed.

Convolutional neural networks (CNNs) are widely used in various image tasks because of their local linkage and translational invariance. In HSIC, CNNs are usually designed from spatial and spectral dimensions. In some works [32], [33], [34], 2-D CNN is used to extract information of spatial dimension and spectral dimension simultaneously for classification of HSIs. Furthermore, in [35] and [36], 3-D CNN is used to extract spectral information with finer granularity. In [37], a spectral-spatial residual network (SSRN) is designed, which uses continuous spectral and spatial residual blocks to extract feature maps from hyperspectral contexts, thus reducing the gradient disappearance phenomenon in neural networks. Zhu et al. [38] designed an end-to-end residual spectral-spatial attention network (RSSAN) for HSIC. It achieves a better classification effect by stacking the spectral attention module and spatial attention module. Xing et al. [39] presented a novel deep network with irregular convolutional kernels and self-expressive property (DIKS) for HSIC. In [40], by combining the attention mechanism with CNN, a method based on the attention mechanism is proposed, called multilevel feature network and spectral-spatial attention model (MFNSAM), which consists of a multilevel CNN and a spectral-spatial attention module.

In recent years, graph neural networks (GNNs) have attracted extensive attention because of their ability to model non-Euclidean data structures. In [9], spectral-spatial graph convolutional network (S^2 GCN) was designed to combine spatial information and spectral information in HSIs for classification. Mou et al. [41] proposed a nonlocal graph convolutional network (nonlocal GCN), in which the edges between nodes are composed of pin-pair similarity, and applied the potential spatial similarity in GCN to HSIC. Wan et al. [42] constructed a multiscale dynamic graph convolutional network (MDGCN) and achieved good classification results in HSIC through superpixel segmentation and multiscale branch networks. Wan et al. [43] developed a dual-interactive GCN (DIGCN) to capture spatial information at different scales through dual-GCN branches. In [10], a deep attention GCN (DAGCN) is designed to extract features, which uses the attention mechanism to calculate the spectral similarity. Wan et al. [44] designed a context-aware dynamic GCN (CAD-GCN), which can gradually refine the graph through the “edge filter” to adapt to the graph convolution. Ding et al. [45] designed the spatial-spectral adaptive probability graph (SSAPG) structure, and combined with GCN, a novel SSAPGCN is proposed. Xi et al. [46] proposed a cross-scale graph prototypical network (X-GPN) to achieve semisupervised high-quality HSIC. Bai et al. [47] proposed a graph attention network with an adaptive graph structure mining (GAT-AGSM), which uses the

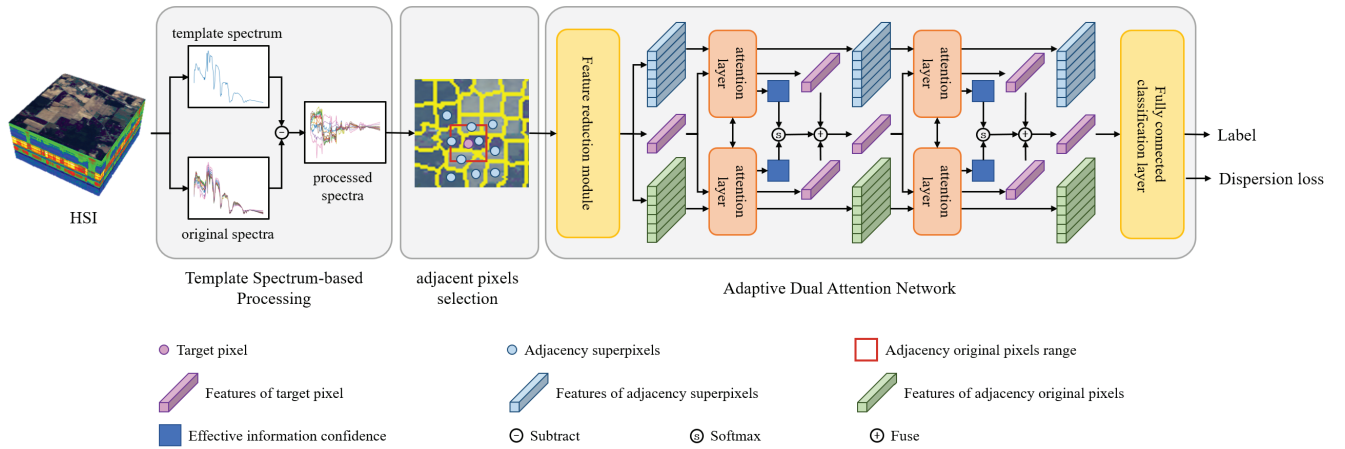


Fig. 2. Overview of the proposed approach for HSIC. The framework consists of a template spectrum-based processing module, adjacency pixel selection module, and ADAN. The ADAN consists of a dimension reduction module, two dual attention layers, and a fully connected classification layer. Dispersion losses are applied to the final fully connected classification layer.

attention mechanism to dynamically generate the unique graph structure of each sample.

With Transformer's [48] widespread success in various tasks, the attention mechanism has received increasing attention. In [49], transformer structure was introduced into HSIC and HSI-BERT was proposed. In HSI-BERT, the sequence of pixels with query pixels and their neighbors is first input into the embedded module with feature and position embedding, and then, the output embedded pixels are input into the BERT module to learn the final feature. The central attention network (CAN) [50] customized the scaled dot product central attention (SDPCA) for HSIs, making the attention mechanism pay more attention to the central target pixel in hyperspectral patch.

III. PROPOSED METHOD

We will introduce the specifics of the proposed approach in this section. Fig. 2 shows the overview of our proposed approach. First, the HSI is segmented by the simple linear iterative clustering (SLIC) algorithm [51] to obtain the superpixel area. To determine the spectrum and coordinates of the superpixel, all pixels in each superpixel area are averaged (spectrum average and coordinates average). After that, the template spectrum is used to process the spectra of pixels and superpixels. Finally, an ADAN performs FE and classification of the target pixel. A feature reduction module, two dual attention layers, and a fully connected classification layer make up the network. The dispersion loss is imposed on the classification layer at the same time.

In the following, we present the template spectrum-based processing module (Section III-A), adjacency pixels selection (Section III-B), the ADAN (Section III-C), and dispersion loss (Section III-D) to describe the crucial module of the proposed approach in more detail.

A. Template Spectrum-Based Processing Module

There are similar trends in the spectral curves of various categories in HSIs due to the overall bias information, which mainly has two influencing factors. The first is the bias

of natural environment and acquisition equipment. In the process of HSI data collection, the setting of data acquisition equipment and natural factors, such as illumination, ambient temperature, air humidity, and atmospheric refractive index, will affect the spectral value. These factors will cause different bands in the spectrum to have different degrees of energy bias and then make the HSI appear the overall bias information. The second is superclass information unrelated to the refined classification task. In the HSI refinement classification task, different refinement categories may belong to the same implicit superclass. For example, in the classification task of different plant categories, the spectra of various refined plant species not only contain their own unique characteristics but also show the common characteristics of the superclass "plant." The features of common superclass are generally stronger than those of refined categories. Common superclass characteristics mask differences of various refined categories and are useless for refined classification tasks. In general, there is unavoidably overall bias information in HSIs that is not helpful for the classification task. This bias information makes the spectra of different categories tend to be similar and the boundaries between different categories become blurred.

We eliminate the influence of redundant information between spectra by extracting and removing template spectrum from HSIs. The template spectrum represents the overall bias information in HSIs and reflects the common trend of spectral curves of various categories. Removing the template spectrum from the original spectrum can effectively expose the unique characteristics of various categories, thus improving the category separability and facilitating the model to distinguish different categories.

We first formalize an HSI dataset. Given an HSI data, denote the set of pixels as $p = \{p_1, p_2, \dots, p_N\}$, c_i^x and c_i^y are the horizontal and vertical coordinates of p_i , and $h_i \in \mathbb{R}^{c \times 1}$ is the spectral vector of p_i , where N is the number of pixels and c is the number of spectral bands.

The following describes the process based on the template spectrum. For the HSI, its spectra are normalized first.

Specifically, each value in the spectral vector subtracts the mean of the vector and then divided by the standard deviation of the vector

$$\text{Normalize}(h^i) = \frac{h^i - \text{mean}(h)}{\text{std}(h)} \quad (1)$$

where $\text{mean}(\cdot)$ and $\text{std}(\cdot)$ are the mean and standard deviation functions, respectively, and h^i is the i th value in the vector h .

Next, the template spectrum of the HSI is extracted. The template spectrum reflects the common trend of the spectra in HSIs and reflects the bias information in different spectral bands. All spectra are arranged into a matrix in which the template spectrum is the main component. Denote $H = \{h_1^T, h_2^T, \dots, h_N^T\} \in \mathbb{R}^{N \times c}$ as the matrix composed of all spectra, and then, the covariance matrix of H is

$$M^C = \frac{1}{c} H H^T \in \mathbb{R}^{N \times N}. \quad (2)$$

Since the covariance matrix M^C is a real symmetric matrix, the eigenvalue decomposition of the matrix can be performed

$$M^C = Q \Lambda Q^T \quad (3)$$

where Λ is a diagonal matrix in which the eigenvalues of M^C are sorted in descending order of value. Also, Q is a matrix in which the eigenvectors of M^C are normalized and arranged in columns, that is, each column is an eigenvector of M^C . The first column of the matrix Q constitutes the transition matrix $P \in \mathbb{R}^{N \times 1}$, and then, the template spectrum is

$$h_t = \text{Normalize}(H^T P) \in \mathbb{R}^{c \times 1}. \quad (4)$$

The template spectrum obtained by extracting the main components of the spectral matrix contains the overall bias information in HSIs. By subtracting the template spectrum, the original spectrum effectively reveals its own unique characteristics. The processed spectrum h' is

$$h' = \text{Normalize}(h - h_t). \quad (5)$$

B. Adjacency Pixels Selection

For ADAN, the input data include target pixel, adjacent original pixels, and adjacent superpixels. The target pixel refers to the pixel that needs to predict the label. The adjacent original pixels and the adjacent superpixels provide effective information to the target pixel. We use p for the original pixels, s for the superpixels, and p_t for the target pixel. In the training process, the label of the target pixel is visible, while the labels of the adjacent original pixels and the adjacent superpixels are hidden.

Superpixels are averaged from multiple original pixels, so they have lower self-variability and larger receptive field, which can provide broader overall local information for target pixel. However, the original pixels retain more detailed features and can provide refined local information for the target pixel. Therefore, the original pixels and superpixels provide spatial information for the target pixel in the neighborhood from the perspective of detail and overall, which can reduce the variability of the target pixel more effectively than using

a single scheme of adjacent pixel, thus improving the classification accuracy.

The adjacent original pixels are the pixel in the patch centered on the target pixel. Given the size r of patch, the adjacency original pixels can be determined by calculating the distance between pixels. The formula for the distance between the original pixels is

$$\text{dist}_p(p_i, p_j) = \max(|c_i^x - c_j^x|, |c_i^y - c_j^y|) \quad (6)$$

where $|\cdot|$ is the absolute value function and c^x and c^y are the horizontal and vertical coordinates of pixels, respectively. Then, the set of adjacent original pixels of target pixel p_t is

$$\text{Set}_t^p = \left\{ p_j \mid \text{dist}_p(p_t, p_j) \leq \left\lfloor \frac{r}{2} \right\rfloor, j \neq t \right\} \quad (7)$$

where $\lfloor \cdot \rfloor$ is a downward integral function.

Adjacent superpixels are the k superpixels closest to the target pixel. The distance between the superpixel and the original pixel is defined as

$$\text{dist}_s(p_i, s_j) = \sqrt{(c_i^x - c_j^x)^2 + (c_i^y - c_j^y)^2} \quad (8)$$

where c^x and c^y represent the horizontal and vertical coordinates of pixels, respectively. Then, the set of adjacent superpixels of the target pixel p_t is

$$\text{Set}_t^s = \left\{ s_j \mid \underset{k}{\text{argmin}}(\text{dist}_s(p_t, s_j)) \right\}. \quad (9)$$

It is worth noting that in order to balance the information provided by the superpixels and the original pixels, the number of adjacent superpixels is set to be the same as the number of adjacent original pixels, that is, $k = r \times r - 1$.

C. Adaptive Dual Attention Network

In this section, we will introduce the structure of the ADAN in detail. The network consists of a dimension reduction module, two dual attention layers, and a fully connected classification layer. Fig. 3 shows the process of obtaining high-level feature vectors of different pixels in a single adaptive dual attention layer.

1) Dimension Reduction Module: Many works have pointed out the existence of redundant information in spectral bands of HSIs. Also, the hundreds of dimensions of the spectrum will bring great computational burden to the subsequent FE. Therefore, a feature reduction module is used to reduce the spectral dimension and, thus, the subsequent computational burden.

The feature reduction module consists of a full connection layer, a nonlinear layer, and a normalized layer. For input x , the output of the feature reduction module is

$$y = \text{LayerNormal}(\sigma(Wx + b)) \quad (10)$$

where $\sigma(\cdot)$ is the Leaky ReLU function whose parameter is 0.2. $W \in \mathbb{R}^{c' \times c}$ and $b \in \mathbb{R}^{c' \times 1}$ are trainable parameters of the full connection layer, where c is the dimension of the input vector and $c' = \lfloor (c/2) \rfloor$ is the dimension of the output vector. This module reduces the spectral dimension of the input data to half of the original.

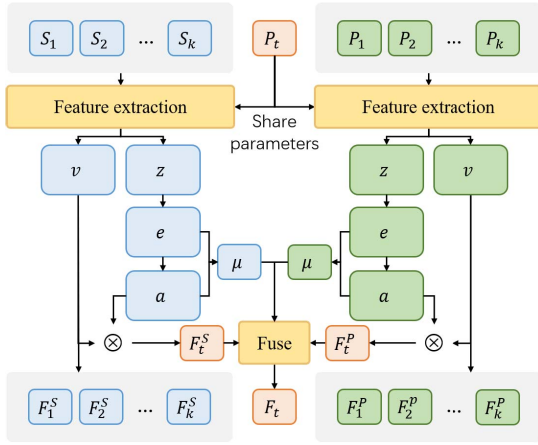


Fig. 3. Process of obtaining high-level feature vectors of different pixels in a single adaptive dual attention layer.

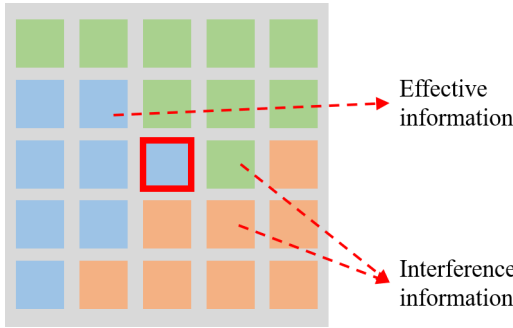


Fig. 4. Schematic of effective and interference information in spatial neighborhood. Squares of the same color represent pixels of the same category, and the square with a red border represents the target pixel.

2) *Effective Information Filtering*: First, it should be noted that all operations in this step are equivalent to adjacent original pixels and adjacent superpixels, so they are collectively referred to as adjacent pixels without specific distinction.

The purpose of adjacent pixels is to provide effective information to reduce the variability of target pixel. Effective information is provided by pixels of the same category as the target pixel, while pixels of other categories provide interference information. Fig. 4 shows a schematic of the effective information and the interference information. When the target pixel combines effective information, its feature vector tends to the corresponding category standard features learned by the model, that is, the variability of its high-level feature decreases. However, there is inevitably interference information in adjacent pixels, which will damage the correct classification of target pixel. Therefore, the target pixel requires selectively combining information in adjacent pixels.

The attention mechanism can effectively reflect the similarity degree between different pixels, and the higher the similarity degree, the more likely the pixels belong to the same category. The attention mechanism usually consists of two parts: attention value calculation and high-level FE. The attention value reflects the degree of similarity between different pixels, while high-level FE mines high-level semantic information in pixels

$$z = W_z h \quad v = W_v h \quad (11)$$

where $W_z \in \mathbb{R}^{c_z \times c}$ and $W_v \in \mathbb{R}^{c \times c}$ are FE matrices and c_z is the length of the vector z .

Since the role of adjacent pixels is to provide effective information for the target pixel, only the attention values between the target pixel and the adjacent pixels need to be calculated, but not between adjacent pixels. Denote z_t as the attention vector of the target pixel, and then, the attention value between the target pixel and the adjacent pixel is

$$e_i = \frac{z_t^T z_i}{\sqrt{c_z}}. \quad (12)$$

The attention value reflects the degree of similarity between pixels. When the degree of similarity is lower than a certain threshold, we believe that the pixel and the target pixel belong to different categories, and the pixel cannot provide effective information for the target pixel. The adaptive threshold θ is

$$\theta = \text{mean}(e) - \lambda \cdot \text{std}(e) \quad (13)$$

where $\text{mean}(\cdot)$ and $\text{std}(\cdot)$ are the mean and standard deviation function, respectively, and λ is the degree coefficient. Then, the set of pixels that provide effective information is

$$\text{Set}_e = \{i | e_i \geq \theta\}. \quad (14)$$

For the target pixel, only the pixel that provides effective information needs to be combined, while for the adjacent pixels, no information from other pixels needs to be combined

$$a_i = \frac{\exp(e_i)}{\sum_{i \in \text{Set}_e} \exp(e_i)}, \quad i \in \text{Set}_e \quad (15)$$

$$h'_j = \begin{cases} \sigma \left(\sum_{i \in \text{Set}_e} a_i v_i \right), & j = t \\ \sigma(v_j), & j \neq t \end{cases} \quad (16)$$

where $\sigma(\cdot)$ is the Leaky ReLU function whose parameter is 0.2.

3) *Dual Feature Fusion*: In order to make full use of the effective information brought by different receptive fields, we designed a dual feature fusion module, which adaptively fused the two high-level features by evaluating the confidence of the effective information in the adjacent original pixels and the adjacent superpixels.

The attention value e reflects the similarity between the adjacent pixels and the target pixel, and the aggregation weight a reflects the degree of contribution of adjacent pixels to target pixel. Therefore, the confidence of effective information is

$$\mu = \frac{\sum_{i \in \text{Set}_e} a_i e_i}{\text{count}(\text{Set}_e)}, \quad i \in \text{Set}_e \quad (17)$$

where $\text{count}(\cdot)$ is the counting function.

The confidence of effective information reflects the amount of information provided by the adjacent pixels to the target pixel. When the similarity between adjacent pixels and target pixel is generally higher, the corresponding confidence degree will be higher. The feature vector of target pixel after adaptive fusion is

$$h = \frac{\exp(\mu_p) h_p + \exp(\mu_{sp}) h_{sp}}{\exp(\mu_p) + \exp(\mu_{sp})} \quad (18)$$

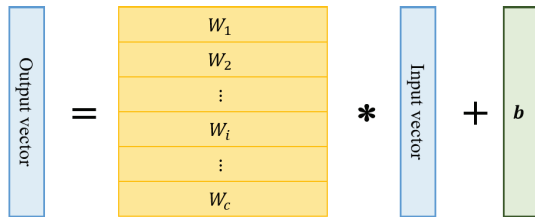


Fig. 5. Schematic of a fully connected classifier.

where μ_p and μ_{sp} are the effective information confidence of adjacent original pixels and adjacent superpixels, respectively, and h_p and h_{sp} are the high-level feature vectors that the target pixel obtains from the adjacent original pixels and the adjacent superpixels, respectively.

Multiple attentional mechanisms help to stabilize the learning process of attentional mechanism. Therefore, the multiple dual attention is also designed. Specifically, M independent dual attention mechanisms were calculated, and their results were concatenated as the final output

$$h = \underset{m=1}{\parallel}^M \frac{\exp(\mu_p^m)h_p^m + \exp(\mu_{sp}^m)h_{sp}^m}{\exp(\mu_p^m) + \exp(\mu_{sp}^m)} \quad (19)$$

where $\parallel$ stands for concatenation and μ^m and h^m represent the effective information confidence and high-level feature vector of the m th attention mechanism, respectively.

D. Dispersion Loss

Neural network models are usually composed of multiple operational layers. For classification tasks, the last layer of neural network is usually a full connection layer, which maps the high-dimensional feature vector to the prediction vector with category number dimensions. Therefore, we can consider the last fully connected layer of the neural network as a linear classifier and all the preceding structures as feature extractors.

In the fully connected layer, for input vector x , output vector $y = Wx + b$, where W and b are trainable parameters. The matrix W can be regarded as a row arrangement of standard feature vectors of multiple categories, and the i th row W_i is the standard feature vector of the i th category. Fig. 5 shows the schematic of a fully connected classifier. The i th position of the prediction vector is

$$y_i = W_i x + b = \|W_i\| \cdot \|x\| \cdot \cos(W_i, x) + b_i \quad (20)$$

where $\|\cdot\|$ represents the norm of the vector and $\cos(W_i, x)$ represents the cosine of the angle between W_i and x .

The process of classification can be regarded as matching the high-level feature vectors of sample with the standard feature vectors of multiple categories. In formula (20), $\|W_i\|$ and b_i are the bias of category, representing the classification preference of the category learned by the model. $\|x\|$ represents the energy of the sample itself, which is the same for different categories and does not affect the relative value of each position in the prediction vector. $\cos(W_i, x) \in [-1, 1]$ represents the matching degree of the sample feature vector and the category standard feature vector. The more similar the

two vectors are, the more likely it is that x belongs to the corresponding category.

In classification tasks, a cross-entropy loss is usually used to measure the gap between the prediction label and the real label and then supervise network model training. The cross-entropy loss is given as follows:

$$L_c = -\frac{1}{N} \sum_i \sum_{q=1}^Q I_{iq} \log(u_{iq}) \quad (21)$$

where N is the number of samples and Q is the number of categories. I_{iq} is the indicator function, if the category of the i th sample is q . It takes 1; otherwise, it takes 0. u_{iq} is the predicted probability that i th sample belongs to the q th category.

It can be seen that the cross-entropy loss is only determined by the component corresponding to the real label in the prediction vector. In other words, the cross-entropy loss is essential to improve the probability of correct classification by increasing the matching degree of the sample's feature vector and the standard feature vector corresponding to the real category. There is no interaction between the sample feature vectors and the standard feature vectors of other categories.

The dispersion loss proposed by us is a supplement to the cross-entropy loss. Unlike cross entropy which is driven by data, dispersion loss is applied to trainable parameters of the classification layer. By separating the standard feature vectors of various categories from each other, the dispersion loss reduces the matching degree between the sample's feature vectors and the standard feature vectors of unrelated categories, thus reducing the probability of misclassification.

Specifically, the cosine similarity between standard feature vectors of different categories is

$$a_{ij} = \cos(W_i, W_j) = \frac{W_i W_j^T}{\|W_i\| \cdot \|W_j\|} \quad (22)$$

where $\|\cdot\|$ represents the norm of the vector. By reducing the similarity between standard feature vectors, the matching degree between sample's prediction vectors and the standard feature vectors of unrelated categories is reduced, and the category separability is improved. The dispersion loss is

$$L_d = \frac{1}{Q} \sum (a_{ij}), \quad i \neq j. \quad (23)$$

In summary, the final loss function is

$$L = L_c + L_d \quad (24)$$

where L_c is the classification loss and L_d is the dispersion loss.

By reducing the gap between the prediction vector and the target vector, the cross-entropy loss reduces the distribution range of the categories, thus reducing the overlapping areas between different categories and improving the classification accuracy. The dispersion loss is directly imposed on the learnable parameters of the classification layer, which makes the distribution centers of different categories far away from each other, thus reducing the overlap area between categories and improving the category separability. The combined use of

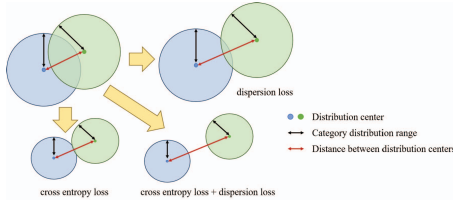


Fig. 6. Distance between samples and distribution centers and the distance between distribution centers of different categories.

cross-entropy loss and dispersion loss can improve the classification separability more effectively, reduce the difficulty of model learning, and improve the classification accuracy. Fig. 6 shows the functions of cross-entropy loss and dispersion loss.

IV. EXPERIMENTS AND RESULTS

Through detailed experiments, we evaluate the effectiveness of the proposed approach in this section. In particular, three widely used HSI datasets are used to compare ADAN and baselines. Four indicators, per-category accuracy, overall accuracy (OA), average accuracy (AA), and Kappa coefficient, are used to verify the performance.

A. Datasets

Three HSI datasets are selected for the HSIC experiments, namely, IP, Pavia University (PU), and Salinas.

1) *Indian Pines*: They were gathered by AVIRIS sensors in the IP forest in northwest Indiana in 1992. The IP data consists of 145×145 pixels. Among the 220 spectral bands of IP, 20 bands are reflected by water. They are 104–108, 150–163, and 220. We generally take the remaining 200 bands after removing these 20 bands as the research object. There are 16 categories in total, but the sample distribution is very uneven.

2) *Pavia University*: The PU dataset was collected in 2003 in Pavia, Italy. The PU data contain 610×340 pixels, and the spatial resolution of each pixel is 1.3 m. The wavelength range of the spectral image is 0.43–0.86 μm . After removing the influence of noise, 103 spectral bands are used to form the image. There are a total of nine categories in the PU dataset.

3) *Salinas*: It was collected by a 224-band AVIRIS sensor over the Salinas Valley in California. Its spatial resolution per pixel is 3.7 m. The Salinas dataset contains 512×217 pixels. Twenty absorbent tapes 108–112, 154–167, and 224 were discarded. It includes 16 categories, such as vegetables, bare soil, and vineyards.

B. Experimental Settings

Our experiment is carried out on a computer with an Intel¹ Xeon¹ Gold 6240 CPU with 128-GB RAM at 2.60 GHz and an NVIDIA GeForce RTX 3090 GPU with 24-GB RAM. The software environment is the Ubuntu 16.04 64 bit and deep learning frameworks of PyTorch.

Based on the datasets introduced in Section IV-A, we set the training rates to 5%, 1%, and 1% on the IP, PU, and Salinas

¹Registered trademark.

TABLE I
NUMBER OF TRAINING AND TEST PIXELS ON THE IP DATASET

Class No.	Class Name	Training	Test	Total
1	Alfalfa	5	41	46
2	Corn-notill	30	1398	1428
3	Corn-mintill	30	800	830
4	Corn	12	225	237
5	Grass-pasture	24	459	483
6	Grass-trees	30	700	730
7	Grass-pasture-mowed	5	23	28
8	Hay-windrowed	24	454	478
9	Oats	5	15	20
10	Soybean-notill	30	942	972
11	Soybean-mintill	30	2425	2455
12	Soybean-clean	30	563	593
13	Wheat	10	195	205
14	Woods	30	1235	1265
15	Buildings-Grass-Trees-Drives	19	367	386
16	Stone-Steel-Towers	5	88	93
Total		319	9930	10249

TABLE II
NUMBER OF TRAINING AND TEST PIXELS ON THE PU DATASET

Class No.	Class Name	Training	Test	Total
1	Asphalt	30	6601	6631
2	Meadows	30	18619	18649
3	Gravel	21	2078	2099
4	Trees	30	3034	3064
5	Painted metal sheets	13	1332	1345
6	Bare Soil	30	4999	5029
7	Bitumen	13	1317	1330
8	Self-Blocking Bricks	30	3652	3682
9	Shadows	9	938	947
Total		206	42570	42776

TABLE III
NUMBER OF TRAINING AND TEST PIXELS ON THE SALINAS DATASET

Class No.	Class Name	Training	Test	Total
1	Brocoli_green_weeds_1	5	41	46
2	Brocoli_green_weeds_2	30	1398	1428
3	Fallow	30	800	830
4	Fallow_rough_plow	12	225	237
5	Fallow_smooth	24	459	483
6	Stubble	30	700	730
7	Celery	5	23	28
8	Grapes_untrained	24	454	478
9	Soil_vinyard_develop	5	15	20
10	Corn_senesced_green_weeds	30	942	972
11	Lettuce_roumaine_4wk	30	2425	2455
12	Lettuce_roumaine_5wk	30	563	593
13	Lettuce_roumaine_6wk	10	195	205
14	Lettuce_roumaine_7wk	30	1235	1265
15	Vinyard_untrained	19	367	386
16	Vinyard_vertical_trellis	5	88	93
Total		319	9930	10249

datasets, respectively. At the same time, the number of training pixels is limited between 5 and 30, and the remaining pixels form the test set. Tables I–III list the numbers of training pixels and test pixels for the three datasets.

Several representative methods were selected for comparison, SVM [30], 2-D CNN [32], 3-D CNN [36], GhostNet [52], RSSAN [38], HSI-BERT [49], MDGCN [42], miniCAN [50], DMSGer [53], and SSFTT [54]. On each dataset, all methods

TABLE IV
OA (%) OF DIFFERENT λ VALUES UNDER EACH ADJACENT RANGE

Dataset	λ	Adjacent Range			
		5*5	7*7	9*9	11*11
IP	0	90.23	91.73	93.15	92.56
	0.5	90.78	93.07	93.79	94.16
	1	90.83	93.30	94.07	94.04
	1.5	92.20	92.46	93.91	94.42
	2	91.78	93.15	94.03	94.08
PU	0	95.08	96.01	96.86	97.58
	0.5	94.38	95.79	96.57	97.40
	1	96.40	96.67	96.05	96.01
	1.5	96.33	96.47	96.64	96.78
	2	96.39	96.20	96.45	96.67
Salinas	0	94.98	96.19	97.91	97.75
	0.5	95.34	96.23	97.22	98.51
	1	94.93	96.62	97.81	98.58
	1.5	95.66	96.58	97.95	98.77
	2	95.65	96.54	98.07	98.45

are trained ten times independently, and the average and standard deviation of the ten results are finally reported.

We use the stochastic gradient descent (SGD) optimizer with a learning rate of 0.01, and the momentum and weight decay are set to 0.9 and 0.0001, respectively. The number of epochs is 500. To prevent overfitting, we use dropout with a parameter of 0.5.

C. Parametric Study

The effective information filtering mechanism in ADAN needs to manually set the degree coefficient λ [see (13)]. The degree coefficient reflects the looseness of the filtering information. The larger the value of λ , the smaller the corresponding θ , the looser the filtering conditions, and more adjacent pixels are selected to provide information. In order to evaluate the influence of the degree coefficient on the classification results of the model, we test the classification accuracy of the model when λ is 0, 0.5, 1, 1.5, and 2 under different adjacency ranges. The adjacent ranges are selected as 5×5 , 7×7 , 9×9 , and 11×11 , and the number of superpixels is the same as the adjacent pixels. The accuracy results on the three datasets are shown in Table IV.

In Table IV, the best accuracy on each dataset has been bolded. Overall, with the expansion of the adjacency range, the classification accuracy is improving. This is because a large range of adjacency can capture more pixels of the same category as the target pixel, which can provide more effective information. When the adjacency range is 11×11 , and the optimal degree coefficient is 1.5 on datasets IP and Salinas, but 0 on PU. This is because the category distribution on the PU dataset is more fragmented (e.g., Asphalt and Trees categories) compared to IP and Salinas. This leads the model to require stronger filtering ability on the PU dataset to remove interference pixels among adjacent pixels. On the IP, PU, and Salinas datasets, the adjacency range of 11×11 was chosen for further experiments, and the corresponding degree coefficients λ were 1.5, 0, and 1.5, respectively.

D. Experimental Results

To verify the performance of the proposed approach, we qualitatively and quantitatively evaluate the classification performance by comparing ADAN with the abovementioned baselines.

1) *Results on the IP Dataset:* The classification accuracy obtained by different methods on the IP dataset is summarized in Table V, where the highest value in each row is bolded. The proposed ADAN outperforms all comparison methods in terms of OA, AA, and Kappa coefficients and achieves the best results. The classification accuracy of ADAN is the best in categories 2, 3, 6, 8, 11, 12, 14, and 16. Compared with RSSAN, HSI-BERT, MDSGer, and SSFTT, the OA of ADAN is improved by 3.18%, 4.36%, 3.32%, and 4.58%, respectively. Fig. 7 shows the classification map generated by different methods on the IP dataset. Fig. 7(a) shows the false-color image of the IP dataset. Fig. 7(b) shows the ground-truth map. It can be seen that compared with RSSAN [Fig. 77(g)], HSI-BERT [Fig. 77(h)], and MDGCN [Fig. 77(i)], ADAN [Fig. 77(m)] has better consistency in category boundary. This reflects that pixels at different positions can obtain effective information to reduce variability in our approach.

2) *Results on the PU Dataset:* Table VI shows the classification accuracy of different methods on the PU dataset, where the highest value in each row is bolded. In terms of OA, AA, and Kappa coefficients, the proposed ADAN achieves the best classification result. ADAN has reached the highest classification accuracy in categories 1, 2, 3, and 5. Compared with the RSSAN, MDGCN, MDSGer, and SSFTT, ADAN improves the OA by 4.08%, 6.45%, 1.23%, and 3.34%, respectively. This once again shows that our method is superior to other comparison methods. Fig. 8 shows the classification map generated by different methods on the PU dataset. Fig. 8(a) shows the false-color image of the IP dataset. Fig. 8(b) shows the ground-truth map.

Note that almost all methods misclassify part of the 3rd category (Gravel) into the 8th category (Self-Blocking Bricks) (the two categories in the middle left). The reason may be that the spectra of these two categories are relatively close and difficult to classify. Apart from that, the classification map of ADAN [Fig. 8(m)] has no other obvious errors compared to the ground-truth map [Fig. 8(b)], which other methods do not.

3) *Results on the Salinas Dataset:* The classification accuracy obtained by different methods on the Salinas dataset is shown in Table VII, where the highest value in each row is bolded. The proposed ADAN has the highest classification accuracy on OA, AA, and Kappa. The classification accuracy of ADAN is the best in categories 7, 8, 9, 10, 12, and 15. At the same time, in the category where ADAN does not achieve the highest accuracy, the gap with the highest accuracy is small. This shows that ADAN can achieve balanced classification results even when the training pixels are unbalanced. This may be the reason why ADAN outperforms the comparison methods in the three indicators of OA, AA, and Kappa.

Fig. 9 shows the classification map generated by different methods on the Salinas dataset. Fig. 9(a) shows the false-color image of the IP dataset. Fig. 9(b) shows the ground-truth map. It can be seen that most comparison methods will

TABLE V
COMPARISON OF EXPERIMENTAL CLASSIFICATION RESULTS ON IP

Class No.	SVM	2D-CNN	3D-CNN	GhostNet	RSSAN	HSI-BERT	MDGCN	miniCAN	MDSGer	SSFTT	ADAN
1	31.46±22.28	53.66±18.75	37.80±12.50	90.73±13.10	97.80±2.14	86.34±15.22	93.17±6.06	78.29±15.94	99.27±2.31	95.61±5.84	97.80±2.14
2	46.39±6.12	64.11±4.71	50.82±6.89	80.80±2.84	87.53±3.58	84.81±4.75	73.07±4.21	75.99±5.56	84.61±4.57	83.95±4.45	90.53±3.12
3	49.31±4.68	70.53±6.03	53.30±6.62	86.51±7.76	94.04±3.53	89.99±4.37	86.46±5.21	75.95±4.92	91.31±4.10	93.69±2.12	94.65±1.65
4	43.20±6.33	58.71±6.14	40.98±8.15	94.80±3.88	86.09±9.74	91.20±8.26	92.58±5.48	71.96±4.79	98.22±2.83	89.73±6.07	92.84±5.98
5	80.83±6.05	88.80±1.89	81.55±6.70	90.39±3.42	96.25±2.11	87.82±5.99	85.86±5.32	87.34±3.27	87.63±5.69	94.92±3.72	94.44±4.16
6	95.06±2.22	97.60±2.03	92.24±5.25	96.24±2.15	98.93±1.13	98.56±0.79	96.53±1.69	96.64±2.32	97.66±1.67	98.66±1.86	99.74±0.29
7	78.26±7.10	98.26±3.04	73.91±15.06	100.00±0.00	99.57±1.37	97.83±3.69	95.22±9.27	90.00±7.29	98.70±2.93	95.65±8.70	95.22±4.78
8	95.51±3.08	98.13±3.59	93.35±4.30	99.16±0.78	99.76±0.69	99.98±0.07	99.36±0.65	99.19±1.43	100.00±0.00	99.80±0.45	100.00±0.00
9	55.33±17.23	95.33±7.06	84.00±13.03	100.00±0.00	100.00±0.00	96.00±5.62	100.00±0.00	86.00±7.57	98.67±4.22	96.67±7.20	96.00±6.44
10	66.63±5.41	69.13±4.56	66.32±7.79	86.37±4.74	92.22±3.47	87.44±3.44	84.55±5.38	85.53±5.90	92.71±4.12	88.40±6.43	92.57±2.38
11	60.48±5.88	61.88±4.41	59.72±5.53	81.02±4.96	86.05±4.38	89.31±2.82	80.23±5.56	73.20±2.71	88.11±2.93	85.13±1.63	95.00±1.26
12	51.51±10.26	69.70±6.11	52.43±8.30	91.85±4.08	94.33±3.23	91.92±3.80	90.02±3.68	80.75±7.09	95.01±3.49	88.13±4.37	95.35±2.08
13	93.74±3.05	97.59±2.91	92.56±7.46	98.26±1.71	98.82±1.76	99.59±0.76	98.10±1.14	99.44±0.15	98.41±2.70	99.79±0.50	99.18±0.50
14	88.78±3.64	94.40±3.25	83.81±5.15	93.51±3.59	98.76±1.15	96.00±2.92	93.58±2.70	96.72±1.91	98.18±1.73	98.79±1.25	99.83±0.17
15	35.67±6.11	79.78±4.27	52.81±11.47	91.17±5.68	93.19±4.83	95.18±2.98	92.02±4.75	76.05±5.01	98.09±3.33	94.69±3.98	96.70±2.74
16	84.43±5.00	90.23±7.23	76.25±9.87	93.64±5.70	95.23±6.47	86.70±13.13	90.57±5.33	84.55±9.05	95.45±5.74	96.59±7.15	99.77±0.72
OA	65.64±0.72	75.13±1.07	66.09±1.54	87.69±1.16	92.24±1.22	91.06±0.42	85.91±1.37	82.60±1.31	92.10±0.49	90.84±0.56	95.42±0.39
AA	66.04±2.20	80.49±1.22	68.24±2.81	92.15±1.09	94.91±1.07	92.42±1.20	90.71±0.84	84.85±1.98	95.13±0.45	93.76±1.27	96.23±0.49
Kappa	60.98±0.86	71.86±1.22	61.62±1.70	86.04±1.31	91.17±1.38	89.81±0.48	84.01±1.53	80.26±1.49	91.03±0.55	89.57±0.64	94.77±0.44

TABLE VI
COMPARISON OF EXPERIMENTAL CLASSIFICATION RESULTS ON PU

Class No.	SVM	2D-CNN	3D-CNN	GhostNet	RSSAN	HSI-BERT	MDGCN	miniCAN	MDSGer	SSFTT	ADAN
1	77.42±8.45	90.48±2.64	79.34±6.61	91.78±3.31	92.21±4.04	83.94±5.28	91.02±3.12	86.71±3.29	92.39±1.66	91.67±3.01	96.96±1.21
2	65.98±5.16	91.19±2.48	85.94±3.62	91.85±4.18	94.12±3.60	95.39±1.32	90.61±2.52	91.61±2.82	97.67±0.68	96.61±1.24	99.29±0.48
3	12.01±10.43	79.93±4.48	66.44±13.38	84.31±6.69	91.07±3.27	73.53±5.11	88.29±5.89	78.34±4.96	98.37±1.28	87.69±3.65	93.94±2.07
4	95.43±2.67	92.03±3.04	88.10±4.13	96.30±1.71	95.65±2.57	92.77±3.18	92.63±1.05	91.43±3.74	95.46±1.40	96.82±1.85	95.83±2.58
5	98.77±0.41	99.35±0.60	91.08±11.70	99.17±1.30	99.21±0.88	99.69±0.39	98.83±0.18	97.79±2.40	99.95±0.09	100.00±0.00	100.00±0.00
6	64.37±7.94	87.51±3.69	70.13±6.39	93.47±2.90	95.17±3.92	90.34±2.35	93.54±1.90	92.60±3.55	99.51±0.94	95.94±3.70	98.95±0.90
7	48.43±31.66	91.88±3.54	75.82±12.97	87.09±5.66	91.64±5.31	71.15±11.34	94.01±5.48	76.76±6.22	97.91±1.98	99.26±0.62	99.01±0.93
8	93.52±5.32	88.41±2.95	77.48±8.59	95.61±2.65	95.04±3.87	87.04±6.28	93.07±2.57	81.07±4.08	97.48±1.87	88.08±4.06	97.02±1.18
9	99.78±0.15	96.56±2.41	96.25±2.36	96.46±3.26	93.40±3.26	82.49±15.49	93.77±6.78	99.16±0.38	93.46±4.16	96.81±1.72	96.58±2.10
OA	70.62±1.89	90.31±0.74	81.61±1.98	92.49±1.63	94.06±1.81	90.15±0.67	91.69±1.15	89.30±0.82	96.91±0.30	94.80±0.85	98.14±0.17
AA	72.86±2.74	90.81±0.57	81.18±2.74	92.89±1.37	94.17±1.18	86.26±2.53	92.86±1.32	88.39±1.00	96.91±0.58	94.76±0.90	97.51±0.41
Kappa	62.80±1.91	87.29±0.91	75.93±2.53	90.17±2.07	92.20±2.32	87.01±0.91	89.14±1.47	85.98±1.03	95.92±0.40	93.14±1.12	97.53±0.22

TABLE VII
COMPARISON OF EXPERIMENTAL CLASSIFICATION RESULTS ON SALINAS

Class No.	SVM	2D-CNN	3D-CNN	GhostNet	RSSAN	HSI-BERT	MDGCN	miniCAN	MDSGer	SSFTT	ADAN
1	97.85±0.83	99.94±0.13	98.50±1.29	99.61±0.47	99.19±2.32	99.92±0.17	97.90±1.80	100.00±0.00	99.97±0.17	99.94±0.16	99.24±0.59
2	98.92±1.12	99.40±0.40	99.56±0.46	99.42±0.47	99.87±0.39	99.99±0.01	99.73±0.36	99.95±0.06	99.25±0.10	99.99±0.02	99.94±0.12
3	90.22±7.62	96.93±3.52	94.38±4.70	98.29±1.46	99.66±0.54	99.36±1.07	99.63±0.62	97.36±2.21	99.99±0.00	99.92±0.19	99.68±0.90
4	98.39±0.73	98.46±1.28	97.96±1.08	98.59±1.26	99.09±1.00	95.41±5.10	98.80±0.80	98.82±1.08	94.89±0.57	99.78±0.37	99.60±0.51
5	95.92±2.13	97.91±1.26	94.77±2.53	96.97±2.57	97.57±1.83	96.19±2.82	97.66±0.62	97.42±1.10	97.89±1.19	98.54±2.02	98.07±1.55
6	99.44±0.78	99.59±0.37	99.46±0.64	99.89±0.24	100.00±0.00	99.33±2.03	99.80±0.00	99.81±0.03	99.62±0.07	99.98±0.06	99.99±0.02
7	99.45±0.03	99.52±0.37	98.54±1.08	99.70±0.24	99.86±0.13	98.55±1.12	99.07±1.49	99.41±0.78	99.48±0.10	99.97±0.08	100.00±0.00
8	64.11±6.23	82.47±3.09	69.73±5.97	88.49±3.20	89.76±1.50	78.83±3.38	87.17±2.54	84.61±2.45	94.42±3.15	88.23±2.58	98.58±0.78
9	98.84±0.17	99.85±0.13	97.29±1.44	98.92±0.72	99.65±0.70	99.76±0.24	99.81±0.23	97.57±2.91	99.80±0.03	100.00±0.02	100.00±0.00
10	87.63±3.81	94.08±2.84	89.77±4.93	95.26±2.26	96.56±1.83	96.67±1.33	96.93±2.01	95.18±1.77	98.54±0.74	98.15±1.48	98.58±1.22
11	88.49±1.16	95.12±2.25	83.20±12.12	96.24±4.32	99.16±1.82	99.09±1.16	98.37±1.99	94.81±2.70	97.71±0.10	99.90±0.18	97.98±1.47
12	99.62±0.34	99.54±1.22	99.71±0.49	99.97±0.04	99.96±0.12	99.22±1.27	98.69±0.73	99.06±0.68	95.89±0.10	99.82±0.26	100.00±0.00
13	98.27±0.66	99.80±0.17	97.96±2.11	98.34±4.48	99.35±0.92	97.55±2.01	93.55±3.57	96.68±1.39	94.40±0.58	98.24±3.08	99.78±0.47
14	89.44±2.73	98.27±0.86	93.83±2.92	98.34±1.86	99.88±0.17	99.21±1.40	95.47±2.60	93.84±2.54	96.99±4.85	96.41±3.50	99.12±0.47
15	59.78±7.74	77.73±4.01	72.67±6.43	86.31±4.22	92.96±2.33	87.00±5.03	92.07±2.35	85.42±3.02	95.23±2.93	91.26±2.50	97.60±1.18
16	92.10±2.54	96.51±1.53	96.20±2.32	97.69±1.91	98.38±2.11	99.93±0.11	99.20±0.56	96.93±2.49	99.79±0.73	98.94±0.94	99.77±0.38
OA	84.63±0.58	92.34±0.47	87.71±1.92	94.80±0.46	96.37±0.40	93.01±0.50	95.41±0.53	93.53±0.77	97.38±0.43	96.02±0.24	99.08±0.26
AA	91.15±0.50	95.95±0.36	92.72±1.33	97.00±0.25	98.18±0.36	96.63±0.25	97.12±0.28	96.05±0.40	97.74±0.32	98.07±0.39	99.25±0.17
Kappa	82.93±0.62	91.48±0.52	86.35±2.13	94.21±0.51	95.97±0.44	92.24±0.56	94.89±0.59	92.80±0.85	97.08±0.48	95.57±0.26	98.97±0.29

confuse the classification between categories 8 and 15 (the two categories in the top-left corner) because the two categories have similar spectral characteristics. However, ADAN

[Fig. 9(m)] can almost perfectly separate the two categories. This is consistent with the category classification results shown in Table VII.

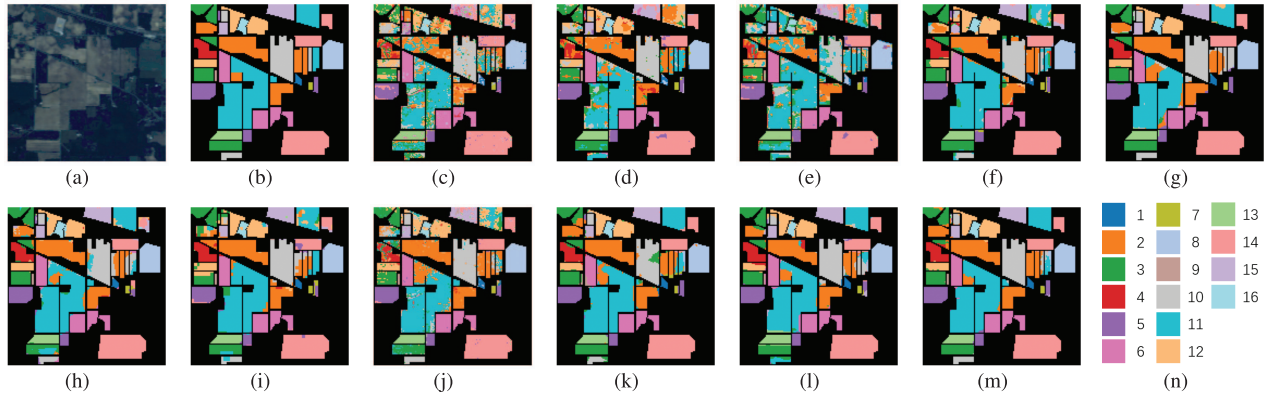


Fig. 7. Classification maps obtained by different methods on the IP dataset. (a) False-color image. (b) Ground-truth map. (c) SVM. (d) 2-D CNN. (e) 3-D CNN. (f) GhostNet. (g) RSSAN. (h) HSI-BERT. (i) MDGCN. (j) miniCAN. (k) MDSGer. (l) SSFTT. (m) ADAN. (n) Category legend.

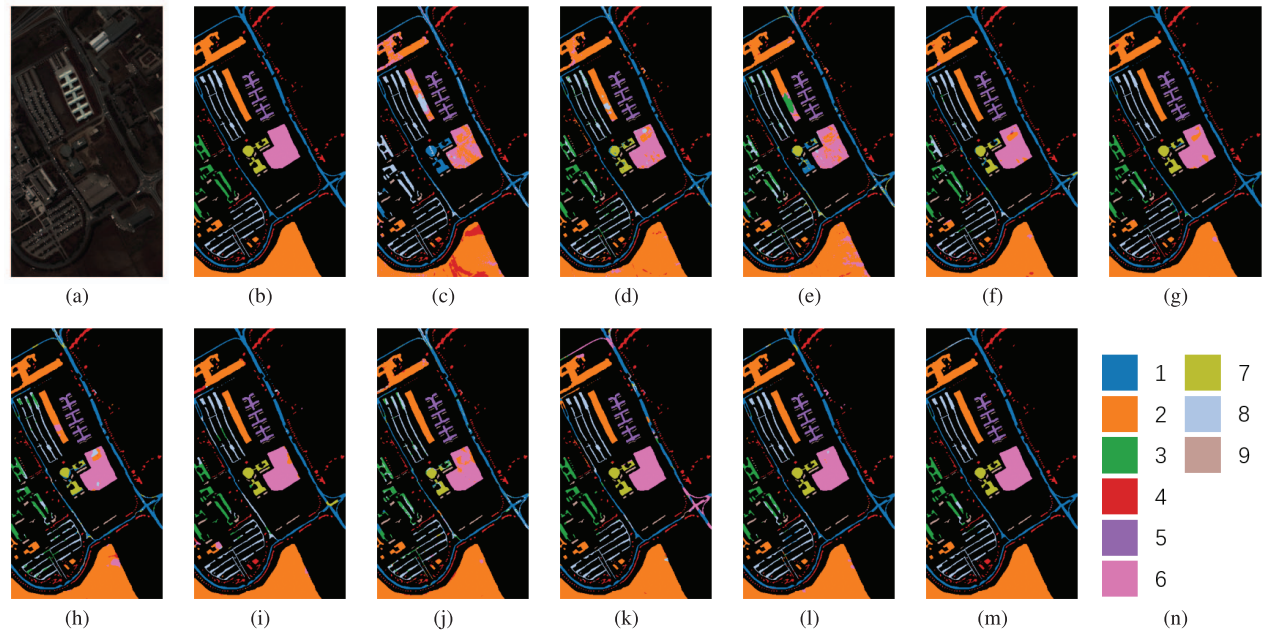


Fig. 8. Classification maps obtained by different methods on the PU dataset. (a) False-color image. (b) Ground-truth map. (c) SVM. (d) 2-D CNN. (e) 3-D CNN. (f) GhostNet. (g) RSSAN. (h) HSI-BERT. (i) MDGCN. (j) miniCAN. (k) MDSGer. (l) SSFTT. (m) ADAN. (n) Category legend.

E. Ablation Study

1) *Effective Information Filtering Mechanism*: To validate the role of effective information filtering mechanisms from adjacent pixels, we designed ablation experiments. When setting the adjacency ranges to 5×5 , 7×7 , 9×9 , and 11×11 , the classification accuracies of the models with and without the filtering mechanism were compared. For IP, PU, and Salinas datasets, the degree parameter λ is set to 1.5, 0, and 1.5, respectively, as shown in Section IV-C. The comparison results are shown in Table VIII.

For the IP and Salinas datasets, in each case of adjacency, using the filtering mechanism is more accurate than not using it. For the PU dataset, in the adjacency range of 5×5 , the filtering mechanism is slightly lower, but in other cases, the classification accuracy using the filtering mechanism is higher. Also, as the contiguous range expands, so does the advantage of using the filtering mechanism.

This may be because the category distribution in the IP and Salinas datasets is relatively regular, and there are enough pixels in the contiguous range that are the same category as the target pixel for calculating appropriate thresholds, while for the PU dataset, the category distribution is more scattered and fragmented. When the adjacency range is small, the filtering mechanism confuses the variability of pixels of the same category and the difference of various categories so that the accuracy of eliminating interference information is insufficient, which in turn damages the classification accuracy. With the continuous expansion of the adjacency range, it is easier for the filtering mechanism to obtain the boundary between effective information and interference information, thereby improving the classification accuracy.

In general, the filtering mechanism can effectively eliminate interference information in adjacent pixels, thereby helping the target pixel aggregate useful information for itself and improve the classification accuracy.

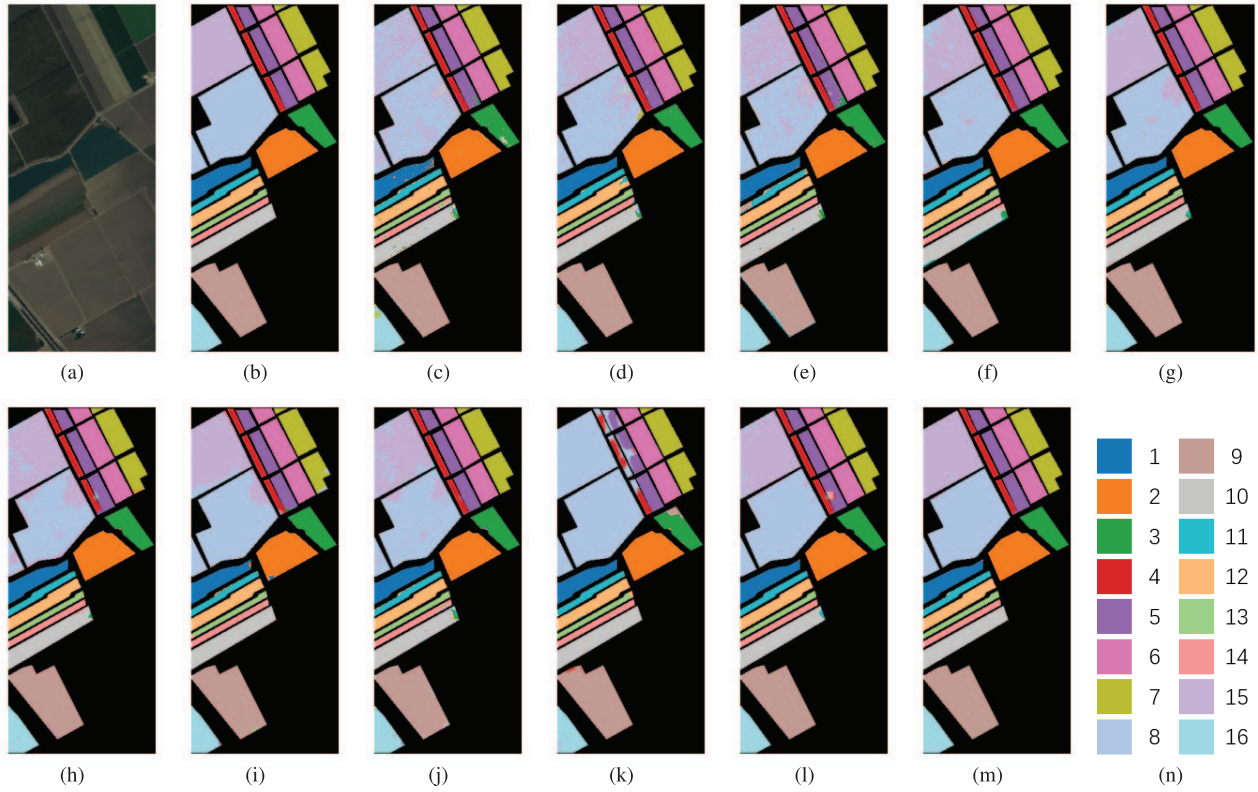


Fig. 9. Classification maps obtained by different methods on the Salinas dataset. (a) False-color image. (b) Ground-truth map. (c) SVM. (d) 2-D CNN. (e) 3-D CNN. (f) GhostNet. (g) RSSAN. (h) HSI-BERT. (i) MDGCN. (j) miniCAN. (k) MDSGer. (l) SSFTT. (m) ADAN. (n) Category legend.

TABLE VIII
OA (%) OF WHETHER TO USE FILTERING MECHANISM IN
DIFFERENT ADJACENCY RANGES

Dataset	Adjacent Range	5*5	7*7	9*9	11*11
IP	no-filtering	90.54	90.88	92.86	91.99
	filtering	92.20	92.46	93.91	94.42
PU	no-filtering	95.33	95.94	96.57	96.76
	filtering	95.08	96.01	96.86	97.58
Salinas	no-filtering	94.66	96.24	97.01	97.72
	filtering	95.66	96.58	97.95	98.77

2) Qualitative Analysis of Template Spectrum-Based Processing Module: In this section, we qualitatively and intuitively analyze the role of template spectrum-based processing module. The spectral curves of the three datasets are shown in Fig. 10. It can be seen from Fig. 10(a), (d), and (g) that different datasets have the phenomenon that the spectral curves of various categories are similar. This is caused by both natural environment factors and superclass. Fig. 10(b), (e), and (h) represents the template spectrum extracted from the three datasets. The template spectrum can effectively represent the overall spectral curve trend of the dataset. Fig. 10(c), (f), and (i) shows the spectral curves of each category after processing on the three datasets. It can be easily seen that, compared with the original spectral curves, the processed curves of various categories are significantly different, that is, more unique spectral characteristics of category are displayed. Such spectral curves that fully expose their own unique

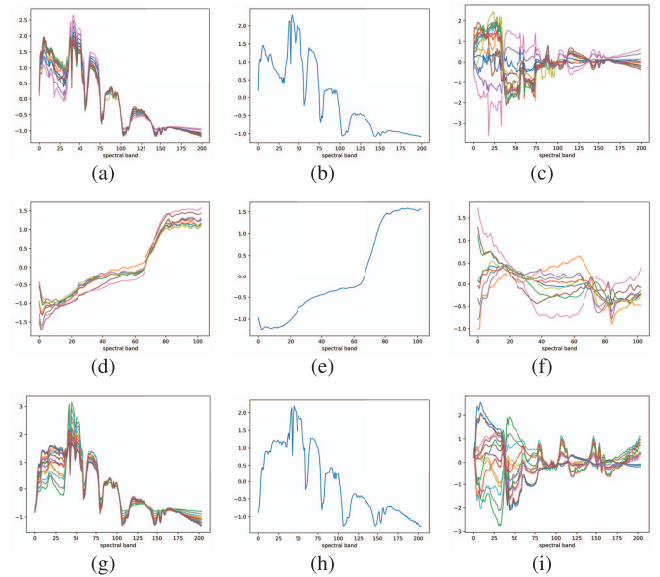


Fig. 10. Spectral curves of three datasets. Different curves represent the average spectrum of different categories. (a), (d), and (g) Original spectral curves of IP, PU, and Salinas, respectively. (b), (e), and (h) Template spectral curves of IP, PU, and Salinas, respectively. (c), (f), and (i) Spectral curves after processing of IP, PU, and Salinas, respectively.

characteristics can effectively increase category separability, reduce the difficulty of extracting high-level semantic features by the model, and then reduce the difficulty of distinguishing various categories.

TABLE IX
PERFORMANCE INDICATORS IN DIFFERENT SETTINGS

No.	Settings					IP			PU			Salinas		
	Template	Spectrum	Pixels	SuperPixels	Dispersion Loss	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
1			✓			78.42	84.06	75.47	92.79	94.18	90.47	92.04	95.51	91.15
2				✓		81.55	74.05	79.00	91.90	93.20	89.35	95.25	95.27	94.71
3			✓	✓		84.09	84.52	81.87	96.09	96.69	94.84	95.78	97.66	95.31
4			✓		✓	80.38	86.01	77.69	94.49	94.86	92.66	92.48	95.66	91.64
5				✓	✓	83.81	75.97	81.55	92.89	93.96	90.65	96.71	96.57	96.33
6			✓	✓	✓	84.36	88.35	82.20	96.62	97.10	95.53	96.03	97.70	95.58
7		✓	✓			92.35	94.77	91.28	95.02	96.10	93.46	94.79	97.57	94.21
8		✓		✓		91.88	90.92	90.74	93.49	93.61	91.44	98.40	98.51	98.21
9		✓	✓	✓		94.37	95.79	93.57	97.01	97.23	96.06	98.92	99.23	98.80

3) *Module Ablation Experiment*: To verify the role of template spectrum-based processing module, ADAN, and dispersion loss, we design the following ablation experiments. First, experiments 1, 2, and 3 are baselines. Experiment 1 only uses adjacent original pixels to extract information, experiment 2 only uses adjacent superpixels, and experiment 3 uses both adjacent original pixels and adjacent superpixels. Experiments 4, 5, and 6 and experiments 7, 8, and 9 add dispersion loss and template spectrum-based processing module to the baseline. The average results of ten independent experiments are shown in Table IX.

First, comparing experiments 1, 2, and 3 with experiments 4, 5, and 6, it can be seen that the dispersion loss helps to obtain better classification accuracy. It is worth noting that in the case of using both original pixels and superpixels, the improvement of the OA by the dispersion loss is not as good as that of using only the original pixels or superpixels, but it maintains a good improvement on the AA. For example, on the IP data, the dispersion loss is improved by 3.83% on AA when using two kinds of pixels, indicating that the dispersion loss can guide the model to obtain a more balanced classification result for various categories.

Second, comparing experiments 1, 2, and 3 with experiments 7, 8, and 9, it can be seen that compared with the traditional processing, the template spectrum-based processing module can greatly improve the results of the model on the three indicators. Especially on the IP dataset, under the same settings, using a template spectrum-based processing module can improve OA by at least 10.28%.

Finally, comparing experiments 1 and 2 shows that for the IP and Salinas datasets, the classification effect of using only superpixels is better than using only original pixels, while for the PU dataset, using only original pixels works better. This shows that the contribution of adjacent original pixels and adjacent superpixels is different for different datasets. Comparing experiments 1, 2, and 3, it can be seen that compared to using only original pixels or superpixels, using both pixels and superpixels can effectively improve the classification effect of different datasets. For the Salinas dataset, although the improvement of OA is not obvious, the improvement of AA is at least 2.15%, indicating that using original pixels and superpixels at the same time can pay more attention to the categories that are more difficult to classify and have a smaller training number. Similarly, similar conclusions can be drawn

from experiments 4, 5, and 6 and experiments 7, 8, and 9, and will not be repeated here.

V. CONCLUSION

In this article, to achieve better category separability of HSIs, we propose template spectral-based processing and an ADAN with dispersion loss from spectral and spatial perspectives, respectively. Specifically, template spectral-based processing fully exposes the unique spectral features of pixels in different categories by eliminating the template spectrum representing the overall bias from the spectra. Second, in the ADAN, the target pixel extracts effective information from the original pixels and superpixels to reduce its own variability. The dispersion loss increases the difference between the standard eigenvectors of different categories by imposing supervision on the classification layer parameters. The experimental results on three HSI datasets demonstrate that our proposed approach has a better performance than the comparison methods.

In future work, we will focus on exploring the deep mechanism of redundant information generation in HSIs and constructing redundancy removal methods that adaptively fit the characteristics of the dataset.

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